

Chapter 1

Realistic simulation and accurate sky reconstruction

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Introduction

The primary goal of the simulation and reconstruction pipeline is to provide a realistic estimate of the capabilities of the QUBIC instrument. However, increasing the realism of the simulations comes at a significant computational cost. In this chapter, we therefore analyse the impact of the main simulation parameters and their associated trade-offs.

The study is divided into two parts. First, we investigate the balance between simulation realism and computational resources. Second, we identify the parameter choices that maximise the sensitivity of the instrument to the tensor-to-scalar ratio, as quantified by the uncertainty σ_r .

Remark. The study presented in this chapter does not represent an exhaustive analysis of all the simulation parameters. Its goal is to provide a solid basis to rely on for future work.

Personal contribution: Even though parameter studies have been performed in the past, notably by Louise Mousset on N_{sub} and k_{max} during her PhD, no significant results or comprehensive analysis of the full set of simulation and reconstruction parameters had been reported. Performing such a study became an important step once the simulation and reconstruction pipeline reached a sufficiently mature and realistic state. In this context, I led the analysis of these parameters, starting in my second year with a study of the convergence of TODs with respect to N_{sub} . The absence of clear convergence, discussed in this chapter, was a major issue over several months and motivated a thorough review of the entire codebase in an attempt to understand its

origin, representing a significant fraction of my work during this period. An explanation was eventually found through multiple discussions with the APC team working on QUBIC, and the underlying effect was independently demonstrated by Alexandre Huchet using the qubicsoft framework, as well as by myself through the development of a toy simulation model. During my third year, once the software had reached a more stable state, I performed a new analysis of N_{sub} for both simulation and reconstruction, based on the σ_r convergence. A similar study was then conducted for the maximum diffraction order k_{max} , together with an optimisation of the reconstruction hyperparameters.

1.1 Realistic simulation

The realism of the simulated data depends primarily on two parameters: the number of integrated sub-bands N_{sub} and the maximum diffraction order k_{max} . In reality, during data acquisition with the QUBIC instrument, both quantities are effectively infinite, which cannot be reproduced numerically. We therefore seek an optimal choice for these parameters, corresponding to a regime in which the simulated TOD no longer varies when they are further increased. It is important to note that this analysis concerns only the simulation framework: it does not reflect a limitation of the real data acquisition process.

1.1.1 Number of sub-bands

The integration sub-bands number was implemented to approximate the frequency integration in QUBIC bandwidth, as shown in equation 5.11. This approximation is :

$$\begin{aligned}\vec{d} &= \int_{\nu_{\min}}^{\nu_{\max}} \mathcal{H}_{\nu} \vec{s}_{\nu} d\nu \\ &\approx \sum_{i=1}^{N_{\text{sub}}} \mathcal{H}_{\nu_i} \vec{s}_{\nu_i} \Delta\nu_i.\end{aligned}\tag{1.1}$$

To be accurate, N_{sub} needs to be high enough to properly describe the continuous integration. An effective method to verify that is to observe how the simulated TOD evolves with N_{sub} . When we found a value after which TOD has converged, we have the value which properly approximates the integration.

TOD convergence

The first quantity we investigate is the convergence of the TOD, that is, the value of N_{sub} beyond which the TOD no longer changes significantly. A first approach consists of directly comparing the TOD generated with different values of N_{sub} . For simplicity, we consider a sky containing only CMB emission. We then simulate several instrumental operators H corresponding to different values of N_{sub} . To limit the computational cost, only a few time samples are considered. The TOD is computed as :

$$\text{TOD}_{N_{\text{sub}}} = H_{N_{\text{sub}}} s_{\text{CMB}}. \quad (1.2)$$

Selecting a detector at random on the focal plane and plotting its TOD for different values of N_{sub} , we observe that each configuration produces slightly different values at each time sample, as shown in Figure 1.1. This first result indicates that the TOD is indeed affected by the choice of N_{sub} . A convergence trend can nevertheless be observed: the TOD obtained with $N_{\text{sub}} = 4$ differs significantly from the others, whereas those obtained with $N_{\text{sub}} = 64$ and 84 are nearly indistinguishable visually.

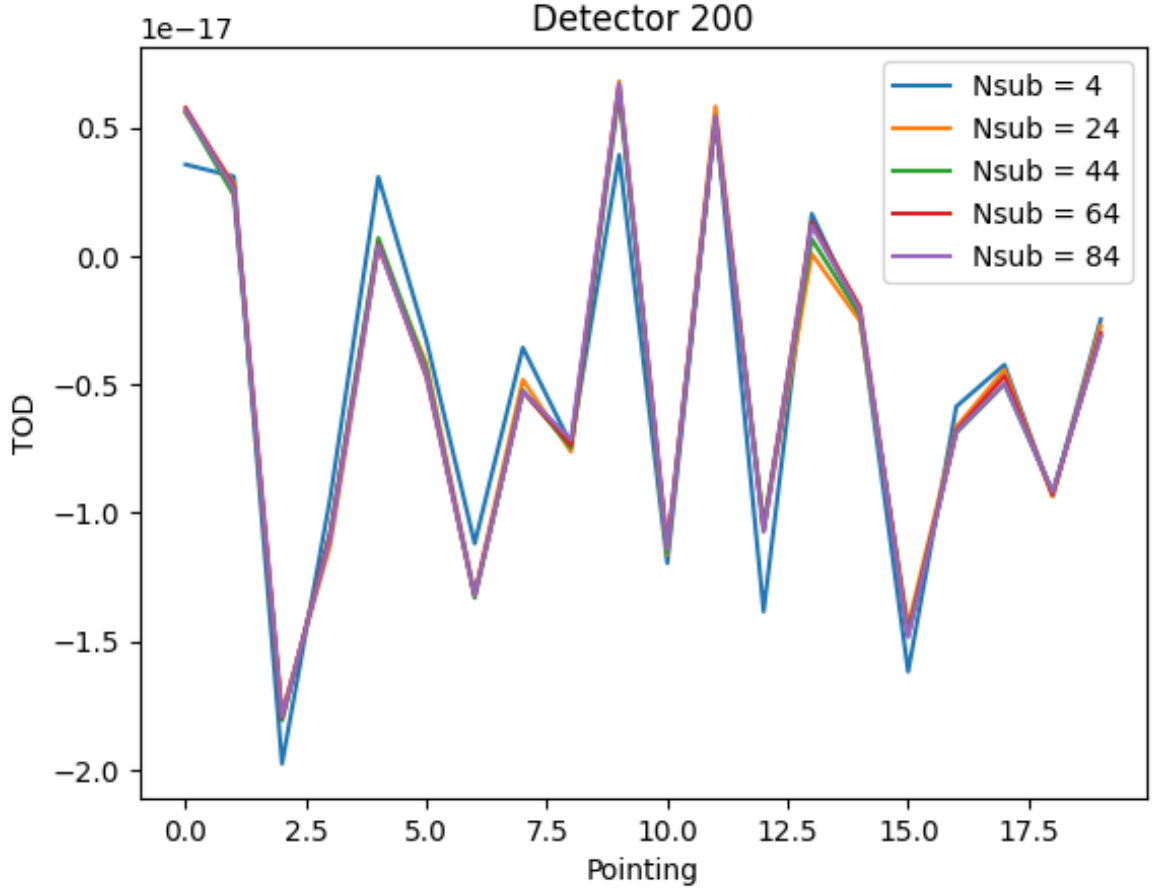


Figure 1.1: TOD for a single detector of the focal plane obtained with different values of N_{sub} .

For a more accurate characterisation, we consider a set of N_{sub} values and compute the corresponding TOD for each case. From these, we define the metric :

$$\delta_i = \frac{\|H(\text{map})^i - H(\text{map})^{\text{max}}\|_2}{\|H(\text{map})^{\text{max}}\|_2}, \quad (1.3)$$

which quantifies the relative difference between the TOD obtained with a given $N_{\text{sub},i}$ and the TOD computed with the largest value $N_{\text{sub}}^{\text{max}}$. The normalisation by the norm of $H(\text{map})^{\text{max}}$ provides a dimensionless estimate of the fractional deviation.

Instead of relying on *qubicsoft*, we developed a simplified acquisition model. For each sub-band, a sky map is convolved with the FWHM of the synthesized beam

peaks. We then compute the pointing direction of each synthesized beam peak for every detector. By summing the contributions from all sub-bands, we obtain a simplified TOD that retains the essential features required to study its convergence with respect to the number of sub-bands. This test was also performed using the full acquisition model implemented in *qubicsoft* to validate the simplified toy model. The results presented below are mainly obtained with the toy model, as it allows the exploration of more complex configurations with a much lower computational cost.

Unfortunately, the TOD does not appear to converge, regardless of the number of integrated sub-bands. The results, shown in Figure 1.2, indicate a relative difference of approximately 5×10^{-3} between two TODs computed using 194 and 204 sub-bands. This difference, as well as the others shown in the figure, is significantly larger than expected, and no clear convergence can be observed.

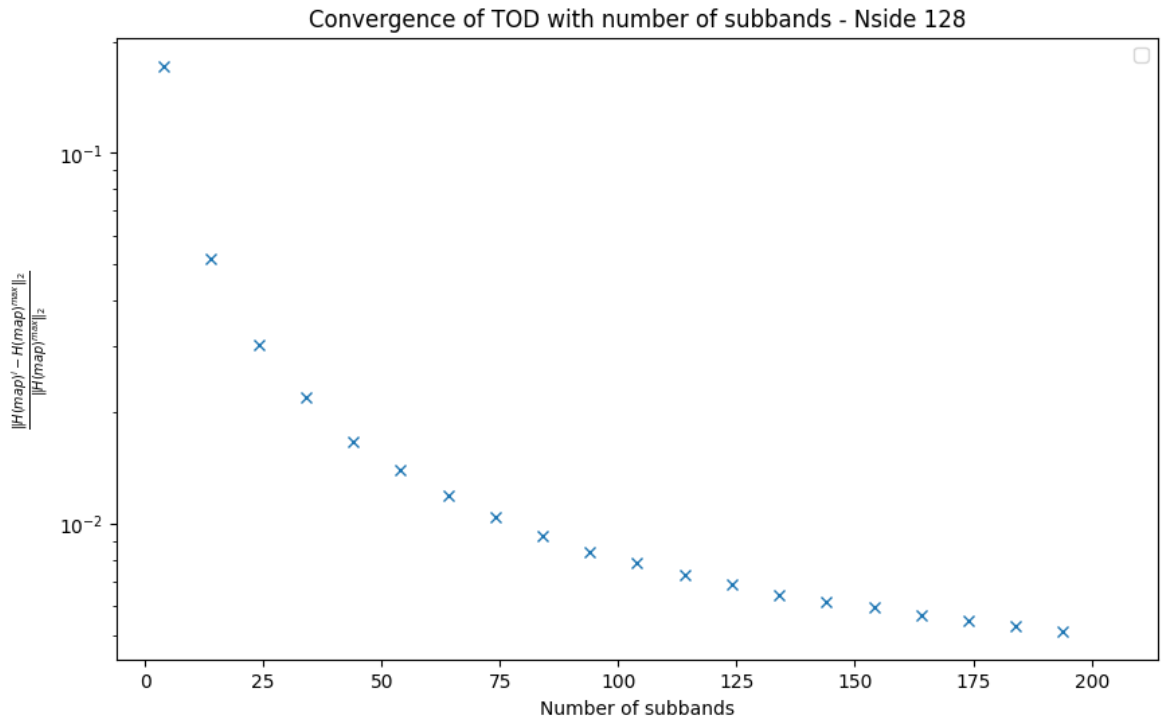


Figure 1.2: TOD convergence with the number of subbands N_{sub} , using *qubicsoft*.

After initially suspecting a bug in the code, which led to several months of testing and verification of each operator composing H , we eventually identified a plausible explanation. Since the sky is represented as a pixelised map, the simulated TOD is affected by pixelisation. Depending on the frequency of the considered sub-band, the synthesized beam peaks do not point to exactly the same positions on the sky. As a result, the sampled pixel value can “jump” whenever a peak crosses the boundary between two neighbouring pixels, even for a very small change in frequency. This effect prevents the TOD from converging completely. This behaviour is illustrated in Figure 1.3, which shows the output of the simplified toy model. For a single time sample, the figure displays the sum of the pixel values sampled by all the synthesized beam peaks as a function of the sub-bands’ frequency. A deliberately coarse map resolution is used to emphasise the effect, which appears as the numerous plateaus

visible throughout the figure.

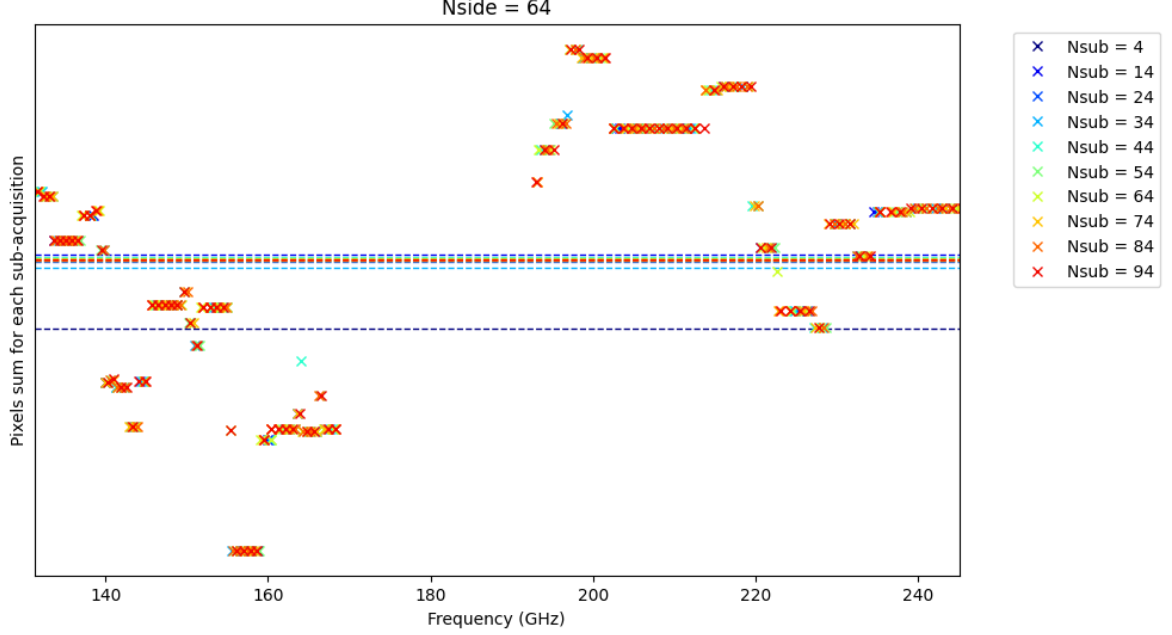


Figure 1.3: Crosses represent the sum of the pixel values sampled by all the synthesized beam peaks for each sub-band, shown for different values of N_{sub} . The dashed lines correspond to the average contribution over all sub-bands.

To confirm that pixelisation is indeed responsible for the lack of convergence, we implemented a pixel interpolation scheme in the toy model to smooth the sky map. Repeating the same analysis with this interpolation yields the results shown in Figure 1.4. In contrast to Figure 1.3, the plateaus disappear and are replaced by a smooth evolution with sub-band frequency. More importantly, the sampled values become essentially independent of the number of sub-bands, providing strong evidence that the observed lack of convergence is primarily caused by pixelisation. Similarly, looking at the dashed lines, representing the average of the contributions from all sub-bands for a given N_{sub} value, they spread less than in the figure without interpolation.

The resulting convergence, with and without interpolation, for a map with resolution $N_{\text{side}} = 256$ and numerous sub-bands, is shown in Figure 1.5. The convergence obtained with interpolation is improved by approximately three orders of magnitude. We initially expected this level of convergence to be achieved without introducing an explicit interpolation, assuming that the beam convolution would already smooth out the map. However, we found that the synthesized beam peaks have a size comparable to that of the HEALPix pixels for $N_{\text{side}} = 256$. As a result, the convolution only partially mitigates the pixelisation effects and has a limited impact on TOD convergence. Moreover, because the synthesized beam FWHM varies with frequency, the convolution itself introduces an additional source of variation between sub-bands, which also

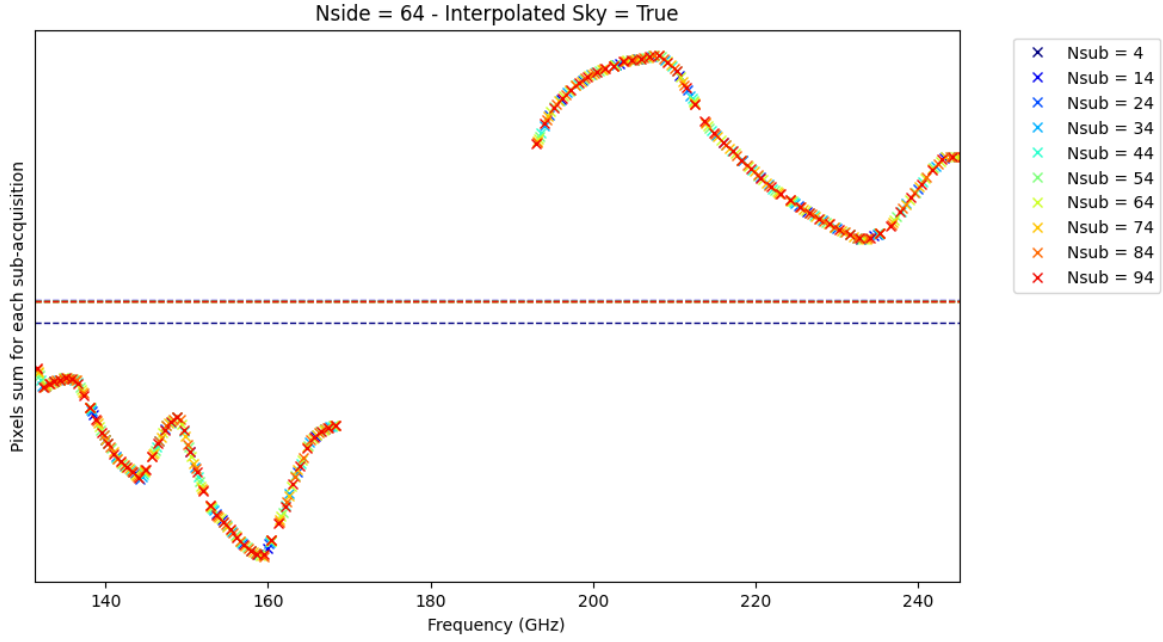


Figure 1.4: Crosses correspond to the sum of pixels' value seen by all the peaks corresponding to a given sub-band, for various N_{sub} numbers, including an interpolated sky. The dashed lines are the average of the contributions from all sub-bands.

affects the convergence of the TOD.¹

After validating this explanation with the toy model, we implemented an optional interpolation operator in the acquisition operator H .

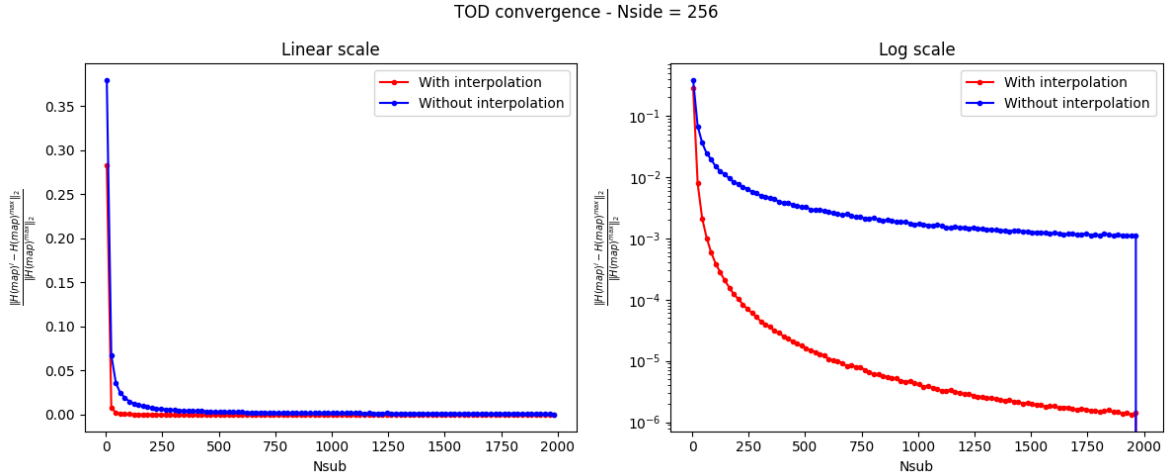


Figure 1.5: Convergence of TOD with the number of subbands, with and without interpolation.

Although these results are satisfying, applying interpolation in the production sim-

¹For completeness, we compared the TOD convergence with and without beam convolution. Somewhat counter-intuitively, the convergence is slightly better without convolution. However, this effect is much smaller than the pixelisation effect and is not investigated further, since omitting the convolution would not correspond to a realistic instrument model.

ulations is not a realistic solution because of its significant computational cost. We therefore investigated an alternative approach to determine a suitable value of N_{sub} for our end-to-end simulations.

Optimal sub-bands number for simulation and reconstruction

Following the results obtained on the influence of N_{sub} on the TOD, we consider an alternative test based on the effect of N_{sub} on σ_r , the uncertainty on the tensor-to-scalar ratio r . This test is, by construction, more involved, as it aims at identifying a realistic value of N_{sub} for TOD generation by studying the behaviour of the uncertainty on a fitted parameter at the end of the reconstruction pipeline. We consider this metric to be particularly relevant, as it directly reflects the final objective of the simulation and reconstruction framework. To perform this analysis, special care must be taken to distinguish the value of N_{sub} used in the simulation from the one used in the reconstruction. In the following, these quantities are denoted respectively by $N_{\text{sub}}^{\text{in}}$ and $N_{\text{sub}}^{\text{out}}$. It is therefore necessary to investigate their respective impact on σ_r , leading to a more complete two-dimensional parameter study.

In many previous simulations (prior to this work), the same value of N_{sub} was used for both simulation and reconstruction, which can introduce an artificial self-consistency between the simulated data and the reconstruction algorithm. In the present work, we avoid this condition in order to improve realism. This study thus provides an opportunity to determine an optimal pair of parameters, both in terms of convergence of σ_r and computational efficiency, including memory usage.

The results of this study are presented in Figure 1.6. The left panel shows the fit using only the two QUBIC frequency bands, while the right panel combines the QUBIC data with the 143, 217, and 353 GHz frequency bands from Planck. In both cases, the figure shows the value of σ_r for each combination of $N_{\text{sub}}^{\text{in}}$ and $N_{\text{sub}}^{\text{out}}$. It is reassuring to observe a similar behaviour in the two configurations. As expected, the cases where $N_{\text{sub}}^{\text{in}} = N_{\text{sub}}^{\text{out}}$ yield the lowest values of σ_r , as the simulation and reconstruction are self-consistent. However, this configuration is not representative of real observations and is therefore not considered further. Excluding these cases, convergence begins to be observed for $N_{\text{sub}}^{\text{in}} \gtrsim 40$ and $N_{\text{sub}}^{\text{out}} \gtrsim 12$. For computational reasons, using $N_{\text{sub}}^{\text{in}} = 80$ is challenging because of the memory requirements. We therefore adopt the combination $N_{\text{sub}}^{\text{in}} = 40$ and $N_{\text{sub}}^{\text{out}} = 16$ for all subsequent end-to-end simulations.

An additional simulation was performed for $N_{\text{sub}}^{\text{in}} = 120$, which was close to the memory limit of the computing centre used for this work. This extra point confirms that the convergence observed for $N_{\text{sub}}^{\text{in}} \geq 40$ is maintained at higher values.

1.1.2 Maximum diffraction order

The other parameter determining the realism of the simulated TOD is the maximum diffraction order k_{max} . This parameter characterises the number of peaks computed within the simulation, following $N_{\text{peak}} = (2k_{\text{max}} + 1)^2$. It is not straightforward to visualise the effect of k_{max} , which is a bit deeper than what it looks like. Given all the figures of the synthesized beam we have shown throughout this thesis, one can easily talk about “central peak” and “secondary peaks” to describe its structure. But, there is no such thing, as the “central peak” appears only for a detector aligned with the optical

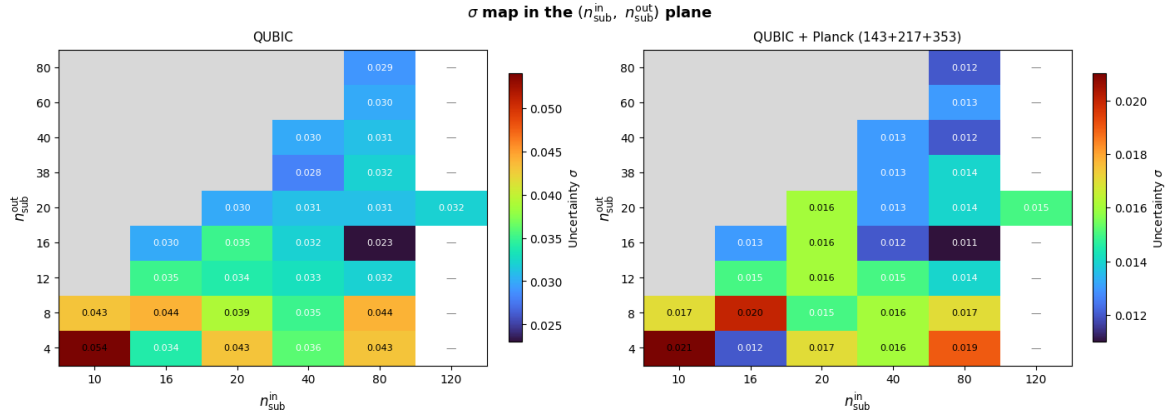


Figure 1.6: Uncertainty on the tensor-to-scalar ratio r , σ_r , for different values of $N_{\text{sub}}^{\text{in}}$ used during the simulation and $N_{\text{sub}}^{\text{out}}$ used during the reconstruction. Left: fit using only the two QUBIC frequency bands. Right: fit combining the QUBIC data with the 143, 217, and 353 GHz Planck frequency bands.

axis of the instrument. In this very specific case, the order 0 peak (when $k_{\text{max}} = 0$, $N_{\text{peak}} = 1$) is located exactly at the centre of the focal plane, where the primary beam is at its maximum. This situation does not happen with the real instrument, because no detector is located at the centre. For an off-axis detector, the brightest peak is not located at the maximum of the primary beam, and is not necessarily the order 0 peak. This situation is illustrated in Figure 1.7, taken from Chapter 4. This can be expanded further : for detectors located far from the centre, an order 2 peak can have a higher amplitude than an order 1 peak. For this reason, it is necessary for the realism to have a k_{max} parameter as high as possible.

Unfortunately, k_{max} is a very costly parameter, the memory increasing quadratically with it. For $k_{\text{max}} = 1$ and $N_{\text{sub}} = 40$, the simulation costs already $\sim 40 - 50$ GB. So, it seems complicated to have a high k_{max} while keeping a realistic N_{sub} . Therefore, we led a similar analysis to show the effect of k_{max} on convergence of σ_r . To avoid using too much memory, a parameter called `synth_beam_fraction` was implemented. This parameter, with value between 0 and 1, corresponds to the fraction of the total synthesized beam amplitude we keep. To give a more practical example, the simulation is computing the amplitude associated with each peak, sort them, and keep the first ones until it reaches the `synth_beam_fraction` value. This method allows us to consider each peak, while saving memory by conserving only the ones contributing the most. This parameter is also included in our analysis. In addition, and similarly as N_{sub} , we have to define the k_{max} used to simulate data, noted $k_{\text{max}}^{\text{in}}$, and the one used for the reconstruction $k_{\text{max}}^{\text{out}}$. For realism, we impose that the `synth_beam_fraction` associated with $k_{\text{max}}^{\text{out}}$ is always equal to 1, while the one for reconstruction can vary.

Unfortunately, memory limitations² prevent us from running the full simulation

²During the writing of this thesis, a method was developed to generate the TOD in chunks, making it possible to simulate observations with much larger values of both $N_{\text{pointings}}$ and k_{max} . However, the reconstruction stage still requires more moderate parameter values. As this implementation became available only at the end of the thesis, it was not used for the present analysis. Nevertheless, it provides a practical solution for performing future studies under more realistic simulation conditions.

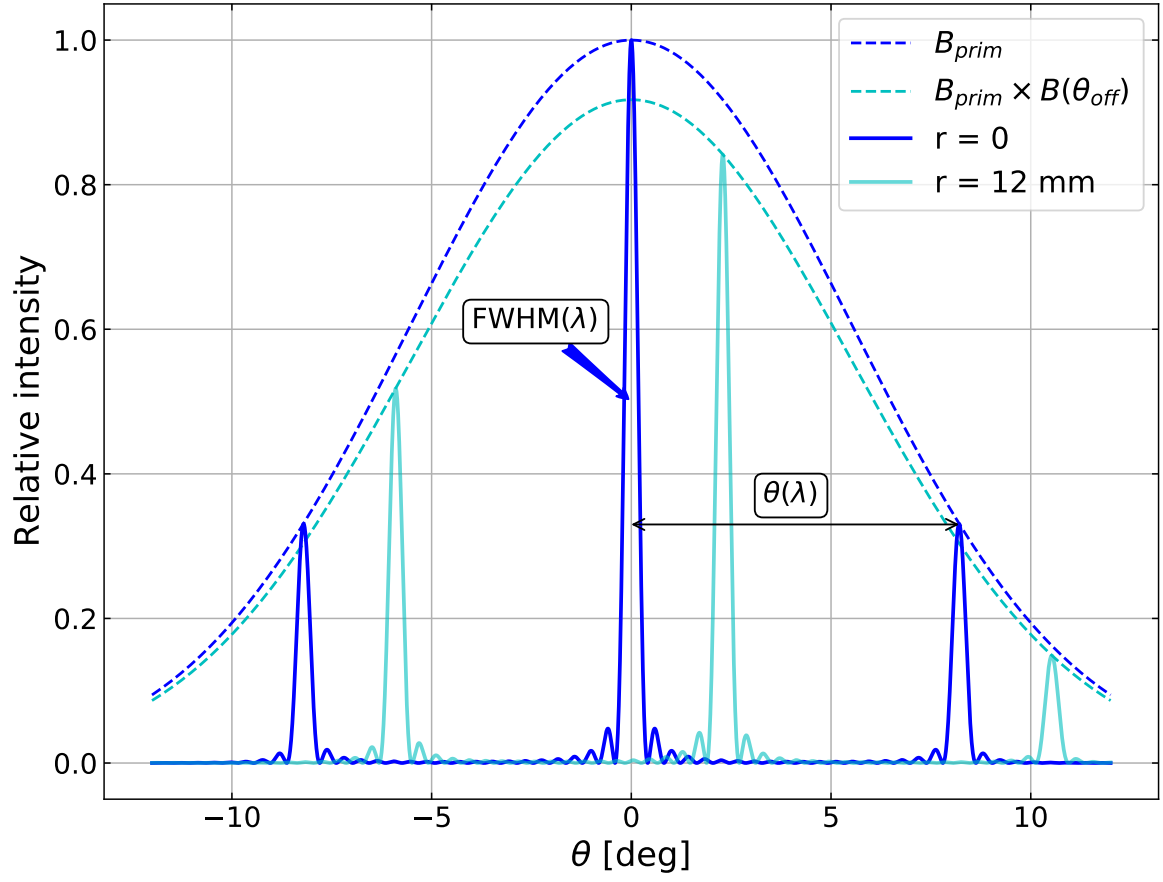


Figure 1.7: 1D cut of the synthesized beam for two detectors: one centred on the optical axis and one off-centre by 12 mm. Taken from [1].

pipeline for $k_{\max} \geq 3$. To get around this limitation, we investigate the impact of the simulation and reconstruction values of $k_{\max}$ while keeping `synth_beam_fraction`=0.99, meaning that synthesized-beam peaks are retained until they account for 99% of the total beam power. For each combination of $(k_{\max}^{\text{in}}, k_{\max}^{\text{out}})$, we compute the corresponding value of σ_r , following the same methodology as for the N_{sub} analysis. The results are presented in Figure 1.8.

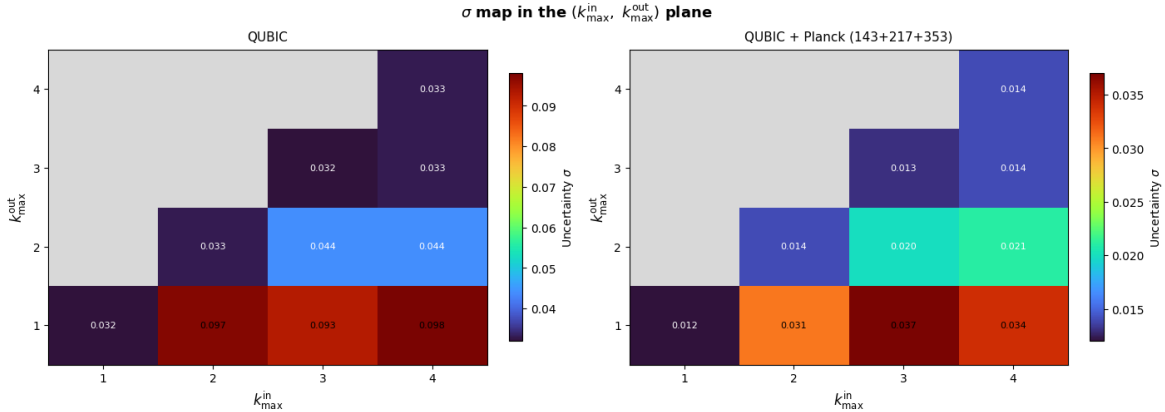


Figure 1.8: Uncertainty on the tensor-to-scalar ratio, σ_r , for different values of $k_{\max}^{\text{in}}$ used during the simulation and $k_{\max}^{\text{out}}$ used during the reconstruction. Left: reconstruction using only the two QUBIC frequency bands. Right: reconstruction combining the QUBIC data with the 143, 217, and 353 GHz frequency bands from Planck.

As expected, the diagonal of the figure provides the lowest values of σ_r , since the same instrumental model is used for both the simulation and the reconstruction. However, it should be noted that, with `synth_beam_fraction`=0.99, all configurations with $k_{\max} \geq 2$ retain the same number of synthesized-beam peaks. Consequently, many parameter combinations lead to identical values of σ_r . A more informative study would therefore require repeating this analysis with `synth_beam_fraction`=1, so that all diffraction orders are explicitly included.

Despite these limitations, the results already highlight a potential weakness of the reconstruction when insufficient memory prevents the use of sufficiently large values of $k_{\max}$. For example, the uncertainty σ_r degrades by more than a factor of three between the configurations $\{k_{\max}^{\text{in}}, k_{\max}^{\text{out}}\} = \{2, 2\}$ and $\{2, 1\}$.

Since the effect of $k_{\max}$ could not be fully characterised, these results should be regarded as preliminary. Nevertheless, they clearly indicate that this parameter deserves further investigation, as an inadequate choice may significantly degrade the reconstruction performance. Because this study remains incomplete, all the results presented in this thesis were obtained using $\{k_{\max}^{\text{in}}, k_{\max}^{\text{out}}\} = \{1, 1\}$. It should, however, be kept in mind that this corresponds to a simplified, and ultimately unrealistic, simulation and reconstruction models.

1.2 Simulation hyperparameters optimisation

1.2.1 Number of time samples : $N_{\text{pointings}}$

The parameter $N_{\text{pointings}}$ corresponds to the number of sky observations, or equivalently the number of time samples acquired during an observation. For a real instrument acquiring data at 100 Hz, a 10-hour observation corresponds to approximately 3.6×10^6 pointings. As explained in Sections 5.3.3 and 5.3.3, running simulations with such a large value of $N_{\text{pointings}}$ is computationally prohibitive.

To overcome this limitation in our simulations, we adopt a random-pointing scanning strategy, in which the instrument points randomly within the observed patch of radius 15° . Compared to a realistic scanning strategy, this approach requires significantly fewer time samples to achieve a homogeneous angular coverage of the observed region, similar to what would be achieved with a very long and optimised scanning strategy. Consequently, a much smaller number of pointings can be used while preserving the statistical properties of the observation, resulting in simulations that remain computationally feasible. This approach corresponds to a “perfect” scanning strategy, which then avoids issues coming from the scanning in our simulation and reconstruction. The noise level is then rescaled to match the expected observation time by defining an effective observation duration. Nevertheless, it remains necessary to determine an appropriate value for $N_{\text{pointings}}$, even within this random pointing framework.

To do so, two complementary tests were performed. The first consists in studying the evolution of the coverage as a function of $N_{\text{pointings}}$. The objective is to obtain a sufficiently uniform coverage of the observed patch. In practice, we seek the value of $N_{\text{pointings}}$ beyond which the normalised coverage no longer changes significantly, indicating convergence.

To illustrate the influence of the number of pointings on the coverage, Figure 1.9 shows gnomonic projections of the QUBIC patch coverage for $N_{\text{pointings}} = 100$ and $N_{\text{pointings}} = 10000$. The left panel corresponds to 100 time samples and exhibits a highly inhomogeneous and asymmetric coverage pattern. In contrast, the right panel, obtained with 10000 time samples, displays a broad Gaussian-like shape with a symmetric pattern and an approximately uniform coverage at a given angular distance from the centre of the patch. This angular uniformity is particularly important, as it ensures that the coverage, and therefore the reconstruction noise, depends only on the angular distance from the patch centre.

The coverage was then computed for various values of $N_{\text{pointings}}$, together with its angular profile and the corresponding angular RMS profile. The objective of this analysis, presented in Figure 1.10, is to determine the number of time samples required for convergence. The left panel shows the angular profile of the normalised coverage. One can observe that its shape converges for approximately $N_{\text{pointings}} \sim 6000$. This result is confirmed by the right panel, which displays the RMS of the normalised coverage as a function of angular distance. The RMS profile stabilises for roughly the same number of time samples, indicating that the coverage has reached convergence.

Remark: One may notice that the maximum value of the angular profile shown in the left panel is not equal to unity, despite the coverage being normalised. This is because each point corresponds to an average within an angular bin. For low values of $N_{\text{pointings}}$, the coverage is highly inhomogeneous, causing the bin-averaged value to differ

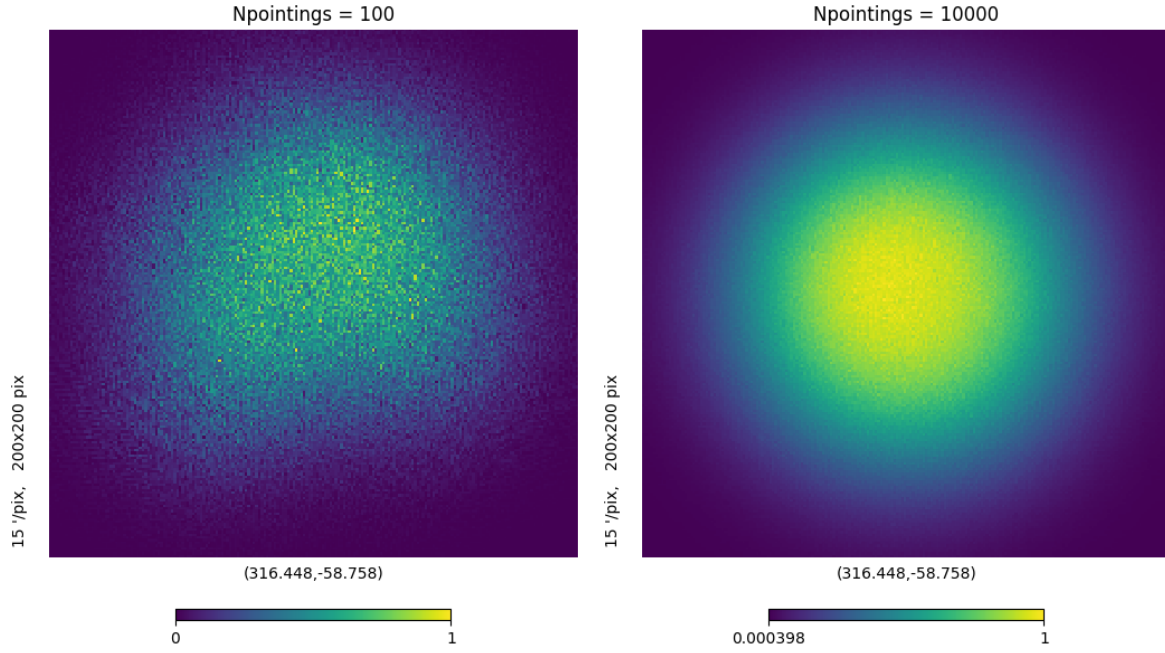


Figure 1.9: Gnomview of the normalised coverage of the observing patch for $N_{\text{pointings}} = 100$ (left) and $N_{\text{pointings}} = 10000$.

significantly from one. The progressive convergence of the peak value towards unity therefore provides an additional indicator of convergence with increasing $N_{\text{pointings}}$.

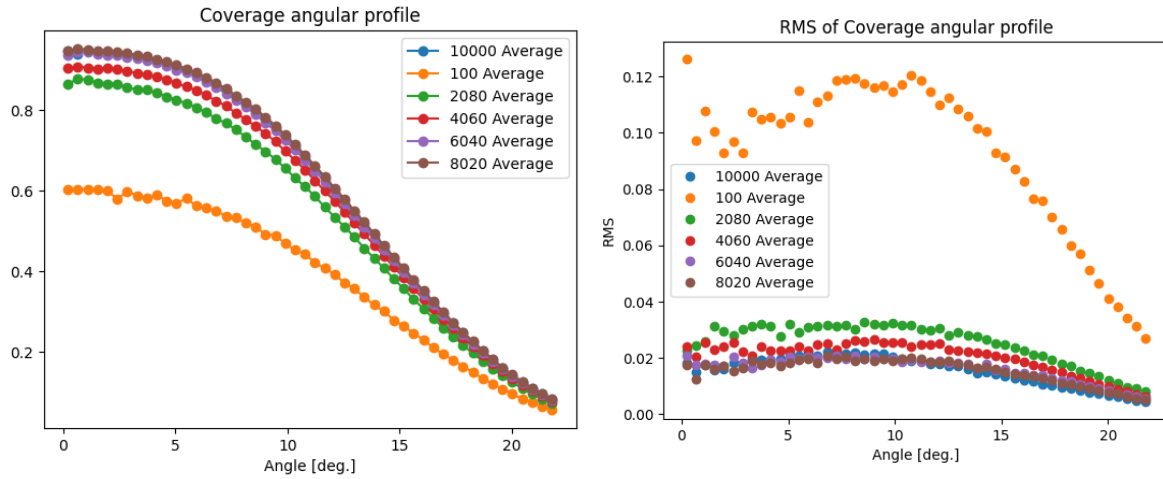


Figure 1.10: Angular profile (left) and RMS of the angular profile (right) of the normalised coverage for various values of $N_{\text{pointings}}$.

A complementary approach consists in studying the convergence of σ_r , the uncertainty on the tensor-to-scalar ratio r , as a function of $N_{\text{pointings}}$. This analysis is performed by running end-to-end simulations for several values of $N_{\text{pointings}}$ and fitting the resulting power spectra (see Chapter 7). The results are shown in Figure 1.11. The lower panel clearly shows that σ_r converges for approximately $N_{\text{pointings}} \gtrsim 3000$.

In light of these two analyses, we adopt $N_{\text{pointings}} = 5000$ for all subsequent sim-

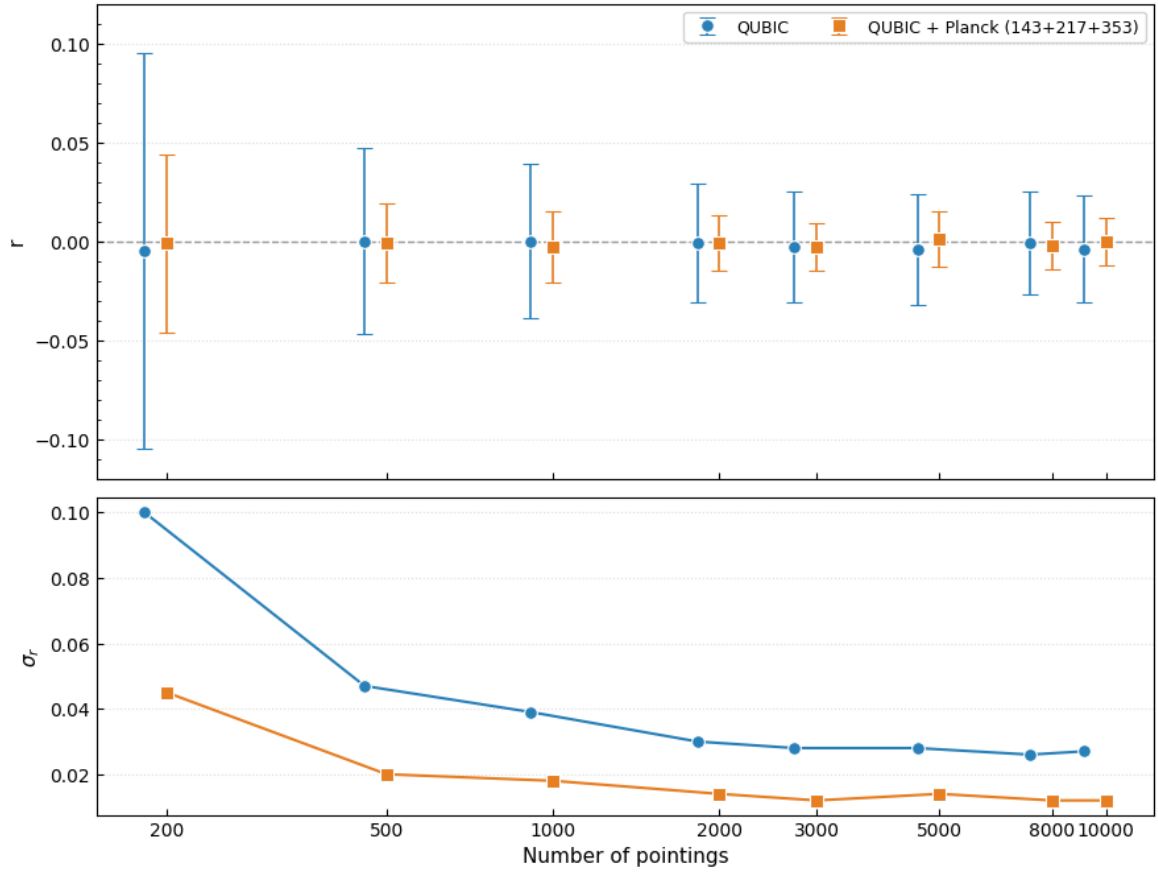


Figure 1.11: Top: fitted value of the tensor-to-scalar ratio r as a function of the number of pointings $N_{\text{pointings}}$. Error bars correspond to the MCMC uncertainty σ_r . Bottom: evolution of σ_r as a function of $N_{\text{pointings}}$.

ulations. This value provides a reasonable compromise between accuracy, memory consumption, and computational cost.

1.2.2 Coverage cut

To improve the signal-to-noise ratio, the outer pixels of the QUBIC sky patch, where the coverage is the lowest, can be excluded from the analysis. This selection is controlled by the *coverage cut* parameter. Defined between 0 and 1, it corresponds to the minimum accepted coverage fraction. All pixels whose coverage fraction, defined as the coverage value at the pixel position divided by the maximum coverage over the patch, is below the chosen coverage cut are removed from the reconstruction.

Applying a coverage cut generally improves the signal-to-noise ratio by discarding poorly observed regions. However, the value should remain as small as possible in order to preserve as many large scales (small ℓ) as possible, as most of the primordial B-modes are there. To determine an appropriate value, we follow the same procedure as in the previous section: we perform simulations for some coverage cut values and study the resulting behaviour of σ_r .

The results on the effect of the coverage cut value are presented in Figure 1.12. A sweet spot is visible for coverage cut equal to 0.2, meaning that the pixels where the coverage is below 20% of the max are discarded. Below this value, σ_r is higher because we are adding pixels with not enough information, leading to a degradation of the signal-to-noise ratio. Above 0.2, σ_r value is rising, which means that we are discarding too much information, and by extension, losing information inside the patch. We are using this value for the end-to-end simulations.

1.2.3 Spectrum parameter : Δ_ℓ

As discussed in Chapter 7, the angular power spectra are computed from $\ell_{\min} = 40$ to $\ell_{\max} = 2 \times N_{\text{side}} - 1 = 511$, with a binning given by the parameter Δ_ℓ . While $\ell_{\max}$ is limited by NaMaster, and $\ell_{\min}$ by the size of the observing patch, it is still necessary to optimise the size of the bins. The binning of the spectra is done for multiple purposes. The main one is, as we observe a small fraction of the sky, NaMaster [2] computes the spectra on a masked map. Doing so, it breaks the orthogonality of the spherical harmonics, mixing power between neighbouring modes. It allows us to have an invertible mode mixing matrix, and therefore to correct for mode mixing. Additionally, having a sky fraction different from 1 results in leakage between E and B-modes, which may produce bias on the tensor-to-scalar ratio measurement. It also helps to improve the signal-to-noise ratio. Indeed, when we are binning, we are averaging multiple modes into one, leading to a smaller covariance (it is divided by Δ_ℓ , see equation 7.12). It also leads to a less costly fitting procedure, as we have to compute fewer cross-spectra values, leading to a runnable MCMC. One can note that binning is possible only because foregrounds' models evolve smoothly in ℓ .

As an empirical and extensive analysis was performed on the spectra parameters in [3, 1], we decided to only test few values to compare with $\Delta_\ell = 30$, found during this previous study. We only ran simulations for $\Delta_\ell = [20, 30, 40, 50]$. The result is visible in Figure 1.13.

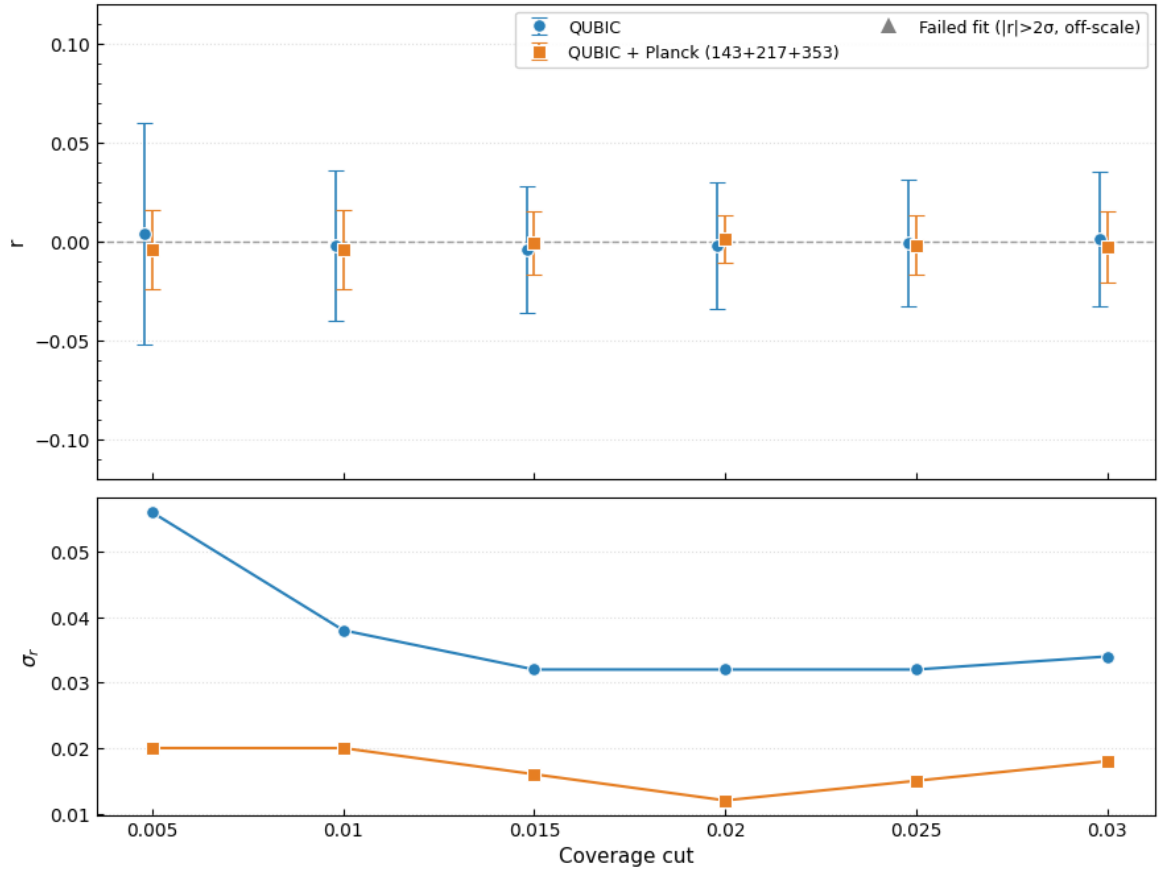


Figure 1.12: Top: fitted value of the tensor-to-scalar ratio r as a function of the coverage cut. Error bars correspond to the MCMC uncertainty σ_r . Bottom: evolution of σ_r as a function of coverage cut.

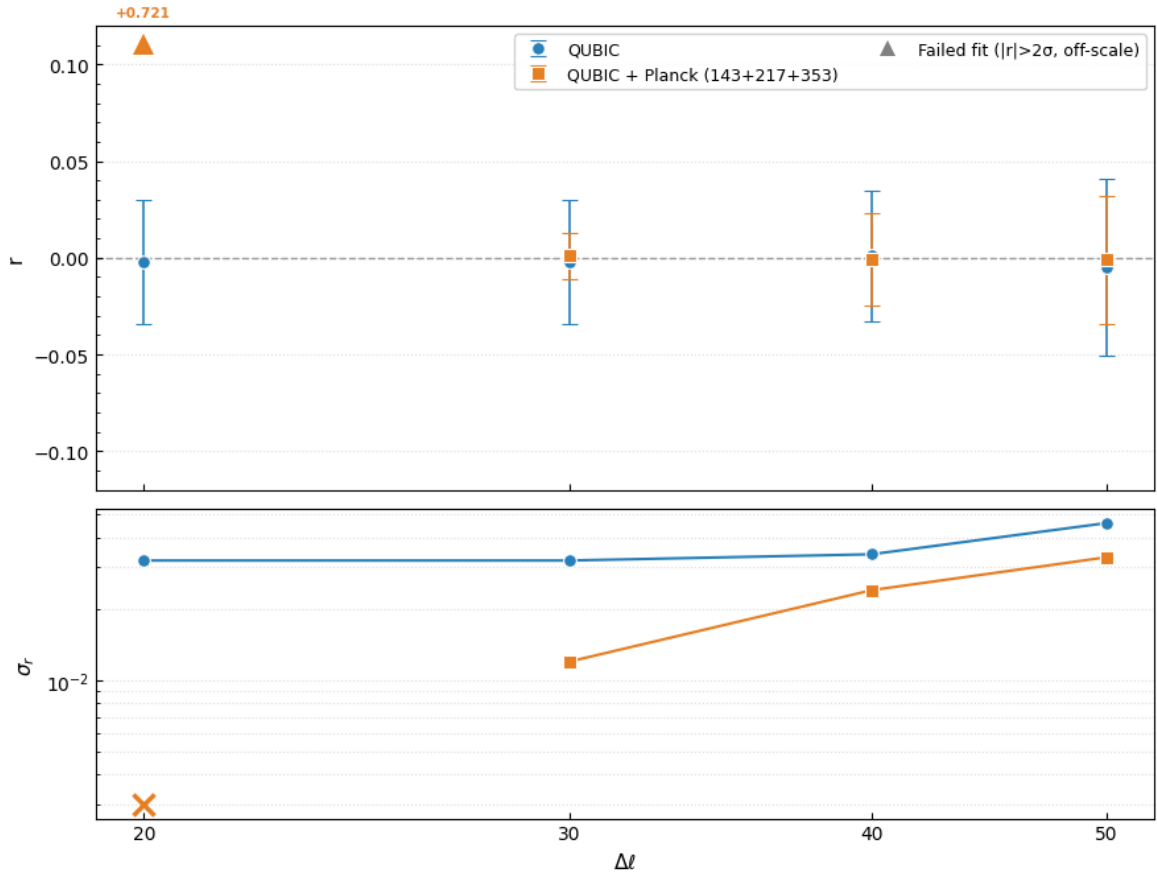


Figure 1.13: Top: fitted value of the tensor-to-scalar ratio r as a function of $\Delta\ell$. Error bars correspond to the MCMC uncertainty σ_r . Bottom: evolution of σ_r as a function of $\Delta\ell$.

If we look only at fit with only QUBIC (blue points), the results are very close. On the other hand, looking at the fit where high frequency bands from Planck are added (orange points), we see that the lowest σ_r is achieved with $\Delta_\ell = 30$, which confirms the previous estimation. But, we see that the fit for $\Delta_\ell = 20$ fails. This behaviour can be understood by recalling the cross-spectrum debiasing procedure described in Chapter 7. For smaller bin widths, the covariance is reduced less efficiently, increasing the sensitivity to residual noise bias. Consequently, a larger number of noise realisations is required to accurately estimate and remove this bias.

Therefore, to avoid the need to increase the already large number of realisations, and as it already shows that best performance is achieved with only QUBIC information, we decided to keep $\Delta_\ell = 30$ for our end-to-end simulations.

1.2.4 CMM convergence parameters : N_{iter} & N_{loop}

In the CMM Chapter 6, we discussed in detail the issues related to the alternating convergence scheme used to reconstruct both the component maps and the mixing matrix (through either blind or parametric models). This algorithm is governed by two parameters: the number of PCG iterations N_{iter} used to reconstruct the component maps, and the number of outer loops N_{loop} . These two parameters must be tuned to achieve optimal convergence in the smallest possible number of steps.

The key parameter is N_{iter} , which largely determines the convergence rate. Indeed, it controls the reconstruction accuracy of the component maps at each iteration of the outer loop. If it is too small, the maps are poorly reconstructed, and the estimation of the mixing matrix becomes biased. If it is too large, the computational cost becomes too important, preventing any practical reconstruction. Moreover, in the most critical case, when the mixing matrix is itself incorrectly estimated (at the beginning of the reconstruction), a large value of N_{iter} can lead the maps to converge towards an incorrect solution, which in turn biases the subsequent estimation of the mixing matrix. Finding the appropriate trade-off is therefore delicate and may depend on the assumed physical model of the sky.

To determine the optimal combination of these parameters, we are considering the convergence of the spectral index, which characterises the reconstruction of the mixing matrix. The analysis is performed by fixing the number of loops to 100 and running the reconstruction for different numbers of PCG iterations per loop³. The starting point is chosen randomly at 2σ from the real value, but is kept the same for all tests. After the loop, a PCG with 150 iterations is run to properly reconstruct the maps, which is not possible with the low N_{iter} values used inside the loop. The number will be increased for the end-to-end simulations. The results are shown in Figure 1.14 and summarised in the Table 1.1. In this table, the first column N_{iter} is the number of PCG iteration per loop, the second column is the loop number where the fit of the spectral index converged, and the third one is the residual value for the spectral index (the difference between the true value and the fitted one).

³Initially, we fixed the total number of PCG iterations and adjusted the number of loops and the number of iterations per loop accordingly. However, since N_{iter} primarily controls the reconstruction of the component maps, whereas N_{loop} governs the convergence of the mixing matrix, we found it more informative to keep one parameter fixed. Moreover, the total number of iterations is not, by itself, a meaningful criterion for assessing convergence.

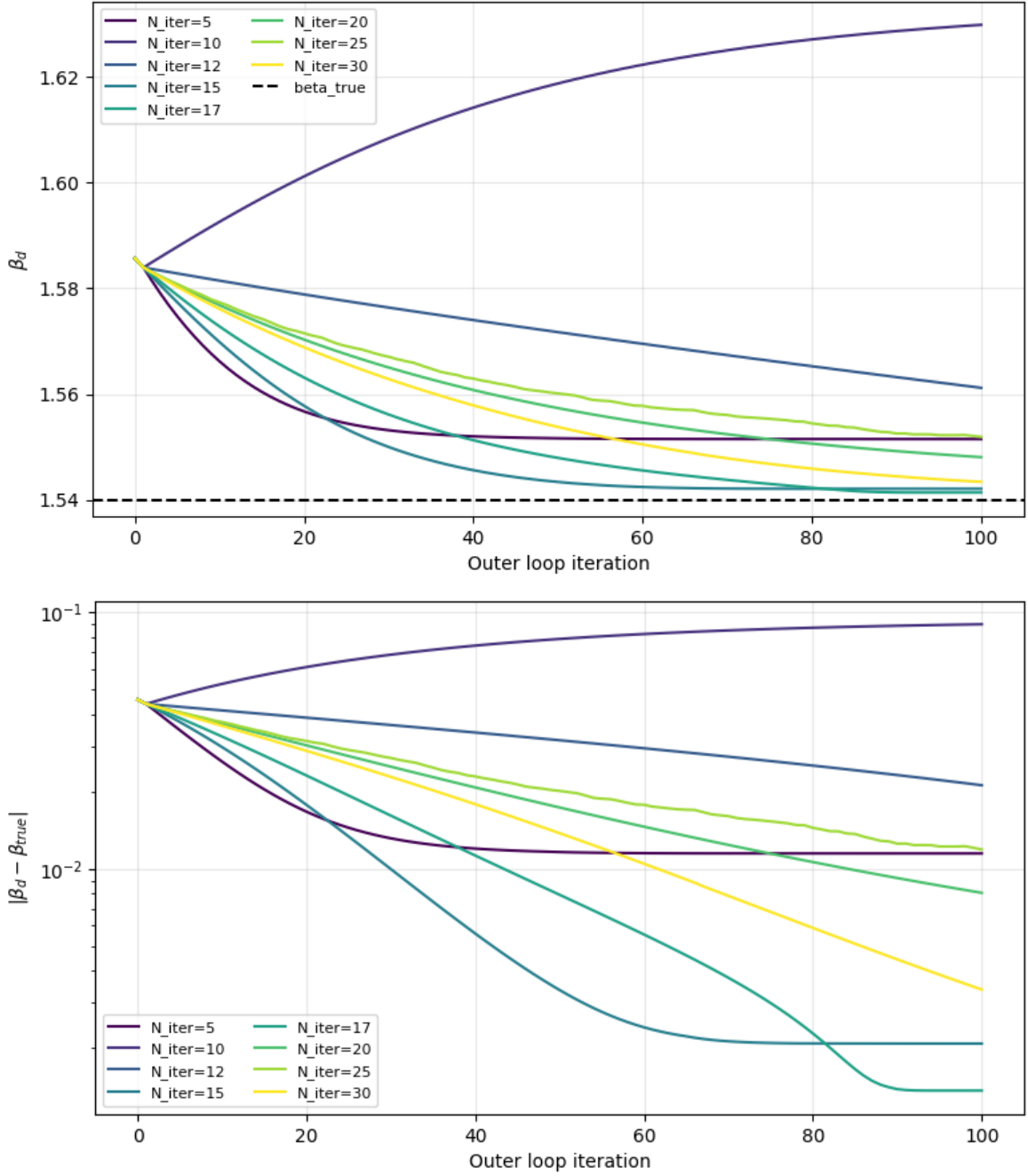


Figure 1.14: Convergence of spectral index β value (top) and residual (bottom) with the number of loop iterations.

N_{iter}	Convergence iteration	β residual
5	52	7.46×10^{-3}
10	$\emptyset$	$\emptyset$
12	> 100	$\emptyset$
15	68	1.35×10^{-3}
17	90	8.86×10^{-4}
20	> 100	$\emptyset$
25	> 100	$\emptyset$
30	> 100	$\emptyset$

Table 1.1: Table summarising the iteration loop number at which convergence is reached and the difference between reconstructed spectral index and true one, for various numbers of PCG iterations within each loop.

The symbol $\emptyset$ in the β residual column means that the fit of the spectral index did not converge, and in the convergence iteration column means that the fit diverges. The tests show that the best value is around $N_{\text{iter}} = 15$, which is the value we will use for the end-to-end simulations as it corresponds to the best trade-off between accuracy and number of loop iterations needed for convergence. The Figure 1.15 shows the convergence of the spectral index for the case $N_{\text{iter}} = 15$.

One would need to be careful with these values, as they do not represent an absolute rule for CMM. Indeed, they are not performed on a large statistical sample but only on one specific case. Furthermore, we have not yet found an explanation for diverging convergence for some N_{iter} values and not for others. An extension of this analysis may be needed in the future, or developing a non-linear solver to remove the need for the alternate convergence.

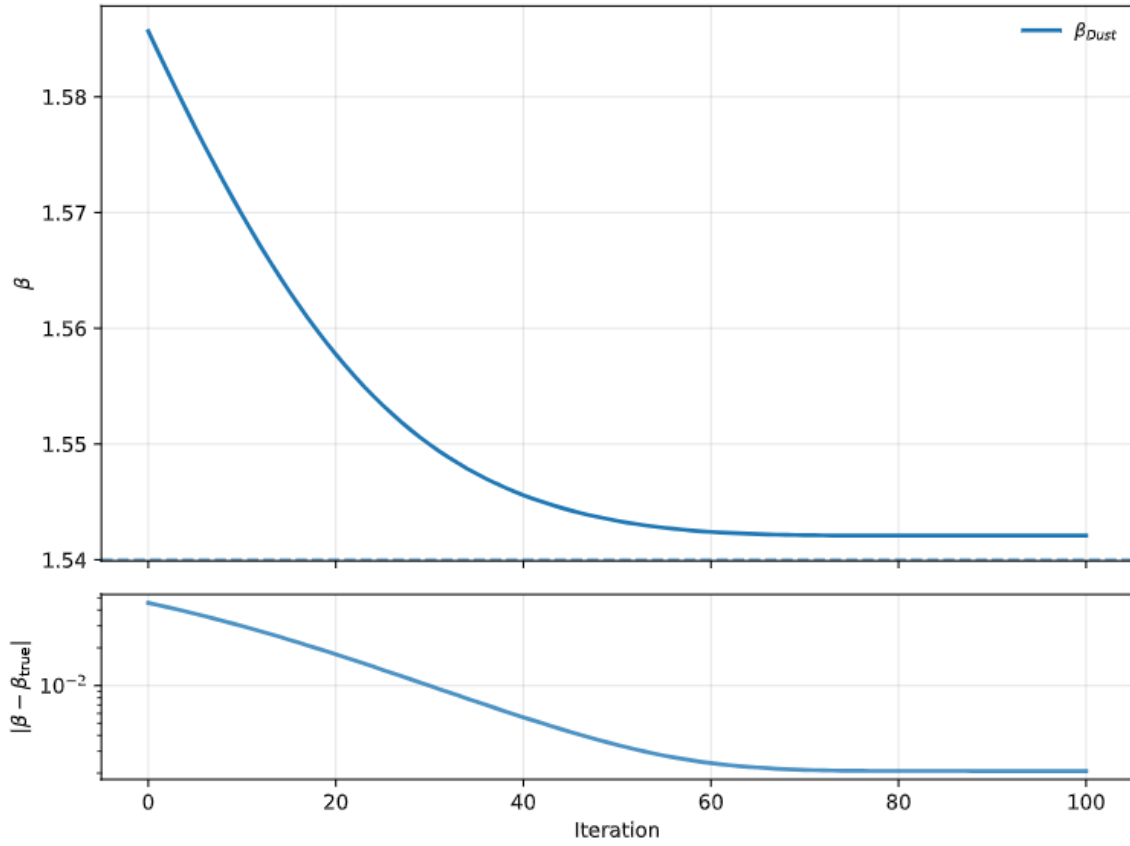


Figure 1.15: Convergence of the spectral index β . The x-axis represents the loop iteration number. (top) Convergence of the β value. (bottom) Convergence of the residual value, the difference between the true parameter and the fitted one.